Statistics

MATH 110: General Education Math

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Statistics

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Populations and Samples

- Define population: the entire group we want to know about.
- Define sample: the part of the population we actually measure.
- Example: population = all adult patients of a Chicago clinic; sample = 50 of those patients whose blood pressure is measured this week.
- Why sample at all: measuring everyone costs too much, takes too long, or is impossible.

Populations and Samples

- Define data: the values we record from the sample.
- A sample is only useful if it looks like the population.
- Example of a bad sample: surveying only patients who come in on weekday mornings (misses people who work days).
- Define random sample: every member of the population is equally likely to be chosen. (Uses "equally likely" from Probability, P3.)

Qualitative and Quantitative Variables

- Define variable: a characteristic recorded for each member of a sample.
- Define qualitative variable: values are categories (blood type, insurance type, smoker or not).
- Define quantitative variable: values are numbers that measure or count (age, temperature, heart rate).
- Test: does it make sense to average it? (Zip codes are numbers but qualitative.)

Qualitative and Quantitative Variables

- Define discrete variable: quantitative, counts (number of ER visits last year).
- Define continuous variable: quantitative, measurements that can take any value in a range (weight, temperature, systolic blood pressure).
- Why it matters: discrete data get one bar per value; continuous data must be grouped into classes; the normal distribution is continuous.
- Sort the variables on a sample patient intake form.

Descriptive Statistics

- Define descriptive statistics: summarizing data with tables, pictures, and a few key numbers.
- The raw data for 20 patients (number of ER visits last year), listed unsorted. Hard to see anything.
- Data set A (keep for the whole unit): 0 visits (6 patients), 1 (7), 2 (4), 3 (2), 4 (0), 5 (1).
- Preview of our tools: frequency tables and histograms, measures of center, measures of spread.

Descriptive Statistics

- Define frequency: the number of times a value occurs.
- Build the frequency table for data set A by tallying.
- Check: the frequencies add to 20, the size of the sample.

Frequency Histograms

- Define frequency histogram: one bar per value (or class), bar height = frequency, bars touch.
- Figure: frequency histogram of data set A.
- Read it back: which value is most common? How many patients had 2 or more visits? $(4 + 2 + 0 + 1 = 7)$

Frequency Histograms

- Continuous data: every value is different, so group values into classes. Define class (an interval of values) and class width.
- Data set B (keep for the whole unit): systolic blood pressure of 40 patients.
- Classes 100–109, 110–119, 120–129, 130–139, 140–149, 150–159 with frequencies 2, 7, 14, 10, 5, 2.
- Figure: frequency histogram of data set B.

Frequency Histograms

- Reading shape. Define symmetric, skewed right (long tail to the right), skewed left (long tail to the left).
- Figures: one of each shape.
- Clinical examples: length of hospital stay is skewed right; data set A is skewed right; data set B is roughly symmetric.
- Define outlier: a value far from the rest of the data.

Experimental Probability

- Recall probability by counting (P3). Define theoretical probability: probability found by counting equally likely outcomes. Coin: $P(\text{heads}) = \frac{1}{2}$.
- Some probabilities have no counting model: will a patient recover full mobility within 6 weeks?
- Define experimental probability: (number of times the event occurred) / (number of trials).
- Example: 210 of the last 250 patients recovered, so the experimental probability is $\frac{210}{250} = 0.84 = 84\%$.

Experimental Probability

- Class activity: everyone tosses a coin 10 times and records the number of heads. Results vary widely.
- Example run: 10 tosses, 4 heads (0.4); 100 tosses, 45 heads (0.45); 1000 tosses, 502 heads (0.502).
- Pool the class results. More trials give an experimental probability closer to the theoretical one.
- Small samples can mislead. (Name the law of large numbers here? Decide.)

Experimental Probability

- Experiment: toss 4 coins and count the heads. Repeat 80 times.
- Results (number of heads 0–4): 6, 21, 27, 20, 6.
- Theoretical probabilities from Binomial Experiments (P8): $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$.
- Expected counts out of 80: 5, 20, 30, 20, 5. Close, not exact.

Relative-Frequency Histograms

- Define relative frequency: frequency / total. It is the experimental probability of that value.
- Data set A: $\frac{6}{20} = 0.30, 0.35, 0.20, 0.10, 0, 0.05$.
- Check: the relative frequencies add to 1 (P2).

Relative-Frequency Histograms

- Define relative-frequency histogram: bar height = relative frequency.
- Figure: data set A, frequency and relative-frequency histograms side by side. Same shape, different vertical scale.
- A patient is chosen at random. $P(\text{2 or more visits}) = 0.20 + 0.10 + 0 + 0.05 = 0.35$ (addition rule, P5).
- $P(\text{fewer than 2 visits}) = 1 - 0.35 = 0.65$ (complement, P6).

Relative-Frequency Histograms

- When every bar has width 1, the area of a bar equals its height. So probability is area.
- Figure: data set A with the bars for 2 or more visits shaded; shaded area $= 0.35$.
- Total area of all bars $= 1$.
- This is the idea that carries over to the normal curve.

Relative-Frequency Histograms

- Figure: relative-frequency histogram of the 80 four-coin experiments (0.075, 0.2625, 0.3375, 0.25, 0.075) beside the theoretical probability histogram (0.0625, 0.25, 0.375, 0.25, 0.0625).
- Experimental histograms approximate theoretical ones, and get closer with more trials.

Relative-Frequency Histograms

- Figure: theoretical histograms for the number of heads in 4, 10, and 20 tosses.
- As the number of tosses grows, the outline of the bars becomes a smooth, symmetric bell.
- Data set B (blood pressure) has the same bell shape.
- Question to leave open: why do such different things make the same shape?

Measures of Center

- Define mean: add the values and divide by how many there are.
- Example: resting heart rates 72, 80, 68, 90, 75. Sum 385, mean $\frac{385}{5} = 77$.
- Symbols: $\bar{x}$ for the mean of a sample, μ for the mean of a population (μ is needed for normal distributions).

Measures of Center

- Mean from a frequency table: multiply each value by its frequency, add, divide by the total.
- Data set A: $\frac{0\cdot 6+1\cdot 7+2\cdot 4+3\cdot 2+4\cdot 0+5\cdot 1}{20} = \frac{26}{20} = 1.3$.
- Same answer from relative frequencies:
  $0(0.30) + 1(0.35) + 2(0.20) + 3(0.10) + 4(0) + 5(0.05) = 1.3$.

Measures of Center

- Define median: the middle value when the data are in order (average of the two middle values if the count is even).
- Heart rates in order 68, 72, 75, 80, 90: median 75.
- Data set A: 10th and 11th values are both 1, so the median is 1.
- Define mode: the most frequent value. Data set A: mode 1.
- The mode is the only measure of center for qualitative data (most common blood type).

Measures of Center

- Lengths of stay (days): 2, 3, 3, 4, 4, 5, 30. Mean ≈ 7.3, median 4.
- One long stay pulls the mean toward the tail; the median barely moves.
- Skewed right: mean $>$ median (data set A: 1.3 vs. 1). Symmetric: mean $\approx$ median. Skewed left: mean $<$ median.
- Figure: the three shapes with mean and median marked.

Measures of Spread

- Two patients, five temperature readings each (degrees Fahrenheit).
- Patient 1: 98.4, 98.6, 98.6, 98.6, 98.8. Patient 2: 97.0, 98.0, 98.6, 99.2, 100.2.
- Both means are 98.6. The center does not tell the whole story.

Measures of Spread

- Define range: largest value minus smallest value.
- Patient 1: 0.4. Patient 2: 3.2.
- Weakness: the range uses only two values, so one outlier changes it completely.

Measures of Spread

- Define standard deviation: a typical distance of the data from the mean.
- Steps: find the mean; find each deviation; square them; average the squares; take the square root.
- Example: heart rates 64, 68, 72, 76, 80. Mean 72; deviations $-8, -4, 0, 4, 8$; squares 64, 16, 0, 16, 64; sum 160; average 32; $\sqrt{32} \approx 5.66$.
- Symbol: σ.

Measures of Spread

- Back to the temperatures: Patient 1, $\sigma \approx 0.13$; Patient 2, $\sigma \approx 1.08$.
- Larger standard deviation means more spread out.
- Figure: two histograms with the same mean, one narrow and one wide.
- Standard deviation is the ruler we use to measure distance from the mean from here on.

Normal Distributions

- Answer the open question: the bell shape appears whenever a value is the result of many small, independent influences (coin tosses added up; heights; blood pressure; birth weights; many lab values).
- Define normal distribution: a symmetric, bell-shaped curve that describes such data.
- Figure: binomial histogram for 20 tosses with the normal curve laid over it.

Normal Distributions

- A normal distribution is completely described by its mean μ and standard deviation σ.
- μ is the center, the peak, and the line of symmetry (mean $=$ median $=$ mode).
- σ controls the width.
- Figures: same μ with different σ; same σ with different μ.

Normal Distributions

- Probability is area (from relative-frequency histograms): the area under the curve between two values is the proportion of the population between them, which is also the probability that a randomly chosen member falls between them.
- Total area under the curve $= 1$. By symmetry, half the area is below μ.
- Figure: normal curve with the area between two values shaded.

Normal Distributions

- Define the empirical rule: about 68% of the area is within 1 standard deviation of the mean, about 95% within 2, about 99.7% within 3.
- Figure: normal curve with the 68–95–99.7 bands marked at $\mu \pm \sigma$, $\mu \pm 2\sigma$, $\mu \pm 3\sigma$.

Normal Distributions

- Systolic blood pressure in a population: $\mu = 120$, $\sigma = 10$.
- 68% between 110 and 130; 95% between 100 and 140; 99.7% between 90 and 150.
- Percent above 130: $\frac{100\% - 68\%}{2} = 16\%$ (symmetry and complement).
- Percent between 120 and 140: $\frac{95\%}{2} = 47.5\%$. Percent below 100: 2.5%.

Normal Distributions

- Reference ranges: a lab's "normal range" is usually the middle 95% of healthy people, $\mu - 2\sigma$ to $\mu + 2\sigma$.
- Example: hemoglobin in healthy adult women, $\mu = 13.5$ g/dL, $\sigma = 1.0$ g/dL. Reference range 11.5 to 15.5 g/dL.
- A result outside the range is unusual: only about 5% of healthy people fall outside it. Unusual does not mean sick.

Z-Scores and Z-Tables

- Define z-score: $z = \dfrac{x - \mu}{\sigma}$, the number of standard deviations x is above (positive) or below (negative) the mean.
- Blood pressure ($\mu = 120$, $\sigma = 10$): 135 has $z = 1.5$; 105 has $z = -1.5$; 120 has $z = 0$.
- A value is unusual if its z-score is above 2 or below -2 (connects to the empirical rule and reference ranges).

Z-Scores and Z-Tables

- Define standard normal distribution: the normal distribution with $\mu = 0$ and $\sigma = 1$.
- Converting to z-scores turns every normal distribution into the standard normal one.
- Figure: blood-pressure curve and standard normal curve, axes aligned (120 with 0, 130 with 1, and so on).

Z-Scores and Z-Tables

- Define z-table: a table giving the area under the standard normal curve to the left of z.
- Excerpt of the table; how to read a row and a column.
- Examples: $z = 1.50$ gives 0.9332; $z = -1.80$ gives 0.0359; $z = 2.00$ gives 0.9772.

Z-Scores and Z-Tables

- Area to the left. Proportion of blood pressures below 135: $z = 1.5$, area 0.9332, so 93.32%.
- Define percentile: the percent of values below a given value. 135 is about the 93rd percentile.
- Figure: curve shaded to the left of 135.

Z-Scores and Z-Tables

- Area to the right (complement, P6). Proportion above 135: $1 - 0.9332 = 0.0668$.
- Area between (addition rule, P5). Proportion between 105 and 135: $0.9332 - 0.0668 = 0.8664$.
- Figures: right-shaded and between-shaded curves.

Z-Scores and Z-Tables

- Working backwards: what blood pressure marks the top 10%?
- Find 0.90 inside the table: $z \approx 1.28$.
- Undo the z-score: $x = \mu + z\sigma = 120 + 1.28(10) = 132.8$.

Z-Scores and Z-Tables

- Birth weights: $\mu = 3400$ g, $\sigma = 500$ g. Low birth weight is below 2500 g.
- $z = \frac{2500-3400}{500} = -1.8$; area to the left 0.0359. About 3.6% of newborns.
- A newborn weighing 4500 g: $z = 2.2$, unusual.
- This is the end goal: from a mean and a standard deviation alone, we can say what fraction of a population lies in any range, and whether a single value is unusual.